

Plant Resilience Score — Project Plan

Revised Edition (incorporates all locked corrections)

Revised: 28 May 2026

1. Executive Summary

A web app for urban plant parents that estimates how well-suited a user's specific environment is for a houseplant. The differentiator is the Plant Resilience Score (PRS): a 0–100 number computed from six hyper-local variables, including indoor-climate modifiers (AC, humidifier) that account for the reality that most urban plants live in artificially controlled environments rather than ambient outdoor conditions.

V1 ships as a working web app with user accounts, multi-plant tracking, and templated advice. V2 adds a paid tier with photo-grounded LLM advice. The entire stack is vendor-neutral — plain Postgres, owned authentication, swappable service modules — so no third party can hold the product hostage.

2. Problem and Differentiator

2.1 The problem

Existing plant-care apps (Planta, Greg, PictureThis) give generic species-level advice. They tell you a 'Pothos' likes indirect light and water every 7–10 days, without accounting for the user living on the 18th floor of a Mumbai apartment with the AC set to 22°C all day. The result is high abandonment — users follow the app's advice, the plant dies anyway, and they conclude they have a brown thumb.

2.2 The differentiator

PRS is a single, interpretable number that quantifies environmental fit. Because the score is broken down by variable, the app can give targeted advice — "Your score is 65, the dominant issue is sunlight. Move 1m closer to your east window."

PRS becomes a retention hook: users return to watch the score change with the seasons, as they move the plant, or as they buy a humidifier. The app turns plant care from a memory task into a measurable game.

3. PRS Methodology

3.1 The six variables

PRS is computed from six independent variables, each normalised to 0–100 before being weighted.

Variable	Source	Notes
Base Hardiness (BH)	Static, per species	Genetic floor. Location-independent by design.
Weather Match (WM)	Open-Meteo + indoor climate inputs	Outdoor data is modified by AC usage, AC setpoint, and humidifier presence before scoring.
Sunlight Access (SA)	User input (4-bucket enum)	bright_direct / bright_indirect / medium / low.

Cross-Ventilation (CV)	User input (4-point scale)	Affects fungal/pest risk and transpiration.
Location Type (LT)	User input (3 options)	Urban / semi-urban / rural. Captures pollution and heat-island effect.
Floor / Altitude (FA)	User input (number)	Modifies microclimate via wind and humidity.

3.2 Formula

Weighted linear sum. Each sub-score is normalised to 0–100, weights sum to 1.0, PRS lands in 0–100.

$$\text{PRS} = (w1 \times \text{BH}) + (w2 \times \text{WM}) + (w3 \times \text{SA}) + (w4 \times \text{CV}) + (w5 \times \text{LT}) + (w6 \times \text{FA})$$

A weighted linear model is chosen deliberately — interpretable, debuggable, and calibratable. Multiplicative or ML models are premature without survival data.

3.3 Weights (V1, mock — not yet evidence-based)

Variable	Weight	Rationale
Base Hardiness (BH)	0.25	Genetics is the strongest single predictor.
Weather Match (WM)	0.20	Drives acute deaths during heatwaves and cold snaps.
Sunlight Access (SA)	0.20	Chronic light deprivation is the #1 indoor killer.
Cross-Ventilation (CV)	0.15	Pests, fungus, transpiration.
Location Type (LT)	0.10	Broad environmental modifier.
Floor / Altitude (FA)	0.10	Microclimate adjustment.

Weights will be re-tuned against real user-survival data once collected. Until then, PRS must not be marketed as scientifically calibrated.

3.4 Indoor climate modifiers (new in this revision)

The original plan calculated WM directly from outdoor Open-Meteo data. This was wrong for indoor plants in climate-controlled apartments. A sealed-AC Mumbai unit runs at the user's setpoint regardless of 35°C outside.

Three new user inputs are collected once per plant and modify the outdoor weather before WM is computed:

- AC usage: Always / Sometimes / Never
- AC set temperature (°C) — collected only when AC usage is Always or Sometimes
- Humidifier present: Yes / No

Logic — Always AC: indoor_temp = ac_set_temp, indoor_humidity = outdoor × 0.5. Sometimes: indoor_temp = (ac_set_temp + outdoor) / 2. Never: indoor = outdoor. Humidifier raises indoor humidity by ~30%.

3.5 Interpretation bands

PRS Range	Status	User message
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80–100	Thriving	Plant is well-matched. Minor optimisations only.
60–79	Stable	One or two interventions will improve health.
40–59	At Risk	Urgent attention. Address the dominant negative variable.
0–39	Critical	Plant unlikely to survive. Relocate, replace, or accept loss.

3.6 Worked example

Pothos in a Mumbai apartment, 12th floor, urban, east-facing window, bright_indirect light, good ventilation. AC always on at 24°C. Outdoor: 32°C / 75% humidity. Indoor (after AC modifier): 24°C / 37.5% humidity.

Component	Calc	Sub-score	Weighted
BH	Pothos baseline	90	22.50
WM	indoor 24°C / 37.5% RH (Pothos range 18–30°C / 40–60%)	85	17.00
SA	bright_indirect	75	15.00
CV	Good	75	11.25
LT	Urban	60	6.00
FA	12th floor	85	8.50
PRS			80.25 (≈ 80)

4. Honest V1 Limitations

Documented openly so an advisor can assess engineering trajectory, not just V1 numbers.

4.1 Weights are not evidence-based

Current weights are directionally informed by horticulture literature. True calibration requires real user-survival data over time.

4.2 Linear model ignores variable interactions

A south-facing high-floor apartment with a Calathea is worse than the sum of its parts (low humidity + high light stress compound). V2 will add multiplicative penalty terms for known killer combinations.

4.3 No per-plant weight overrides activated in V1

Snake Plants do not care about humidity; Calatheas do. V1 uses global weights for all species. The schema includes a nullable `weight\_overrides` JSONB column on `plants`; populated in V2.

4.4 Score history deferred to V2

PRS is computed on demand from live weather. V1 does not store score snapshots — daily snapshots are too noisy to be meaningful. Weekly/monthly trend tracking comes in V2 with proper aggregation.

5. V1 Scope

5.1 In scope

- PRS calculator with all six variables, including indoor-climate modifiers.
- Static plant catalogue of 25–30 popular indoor plants, manually researched.
- Simplified UX inputs — buttons and dropdowns instead of free-form fields.
- Score breakdown — user sees each variable's contribution.
- Targeted templated advice keyed by score band and dominant negative variable.
- User accounts from day one. Email + magic-link authentication, owned in our `users` table.
- Multi-plant tracking. No artificial cap — users can track as many plants as they need.
- Computed watering recommendation (live, weather-driven) displayed on the score page.
- Soft delete with 30-day recovery window. Immediate hard-delete endpoint for GDPR requests.
- Mobile-responsive web app.

5.2 Out of scope (deferred to V2)

- Photo upload and multimodal LLM advice.
- Score history and trend tracking.
- Per-plant weight overrides activated.
- Multiplicative penalty terms for compounding stress.
- Reminders, watering push notifications, browser push.
- Full plant catalogue via external API.

5.3 Explicitly NOT building

- Social features, follows, plant-sharing feeds.
- Marketplace for plants or supplies.
- Community forum.

6. Technology Stack (Vendor-Neutral)

Layer	V1 choice	Notes
Frontend	Next.js	Deployable to Vercel, Netlify, Cloudflare Pages, or any Node host.
Backend	Python (FastAPI)	Deployable to Render, Fly.io, Railway, AWS, or anywhere Python runs.
Database	Plain Postgres	Any hosted Postgres — Neon, Railway, Render, RDS, self-hosted. No vendor-specific features used.
Auth	Owned `users` table	Magic-link email login, implemented in FastAPI. No external auth vendor.
Email	EmailService module	V1 impl swappable — Resend / Postmark / AWS SES / Mailgun.
Weather	Open-Meteo	No API key required. Generous free tier.

Plant catalogue	Static JSON / seeds.sql	Git-tracked, no external dependency.
Storage (V2)	StorageService module	Implementations: S3 / R2 / Backblaze. Vendor chosen at V2 build time.

No layer of the stack ties the product to a single vendor. Every external dependency sits behind an internal interface.

7. Architectural Principles

7.1 Swappable service modules

Every external concern is abstracted: ``AdviceService``, ``PlantDataSource``, ``EmailService``, ``StorageService``. The app code never imports a vendor SDK directly — it imports our abstraction. Swapping vendors is a config change, not a refactor.

7.2 Application-layer authorization

No database-level RLS that ties us to a specific vendor's auth system. Authorization runs in the FastAPI middleware: every endpoint checks ``current_user.id == row.user_id`` before returning data.

7.3 PRS is stateless

Scores are computed on demand. Weather changes hourly; cached scores would be misleading. The DB holds inputs and templates; the backend computes outputs.

7.4 Watering is a computed output

Plants do not have a fixed ``watering_frequency_days`` value. The DB holds ``base_watering_days_summer`` and ``base_watering_days_winter``. Recommended next-watering date is computed live from environment conditions, the same way PRS is.

8. V2 Roadmap

Once V1 has proven retention, V2 introduces paid features.

8.1 Photo-grounded plant advice

User uploads a photo of their plant. A multimodal LLM (vendor selected at V2 build time on cost/latency) generates advice grounded in both the image and the user's PRS breakdown. The ``AdviceService`` signature is additive: ``get_advice(plant_id, prs_breakdown, photo_url=None) → str``.

8.2 Score history and trends

New ``score_snapshots`` table records weekly aggregates. Surfaced as a trend chart on the plant detail page. Foundation for proactive alerts ("Your Calathea has been declining for 3 weeks").

8.3 Per-plant weight overrides activated

The ``weight_overrides`` JSONB column on ``plants`` (already in V1 schema) is populated. Calathea weights WM higher than 0.20; Snake Plant weights it lower.

8.4 Watering reminders and seasonal alerts

Weather-driven push reminders, e.g. "Heatwave starts tomorrow — your Calathea will need watering 2 days early."

9. Next Steps

With math, schema, and architecture locked, the build sequence is:

- Build the static UI front-end (5 screens — Home, Plant Picker, Environment Form, Score Result, My Plants).
- Implement FastAPI backend with the four V1 services (advice, plant data, auth, email).
- Wire Open-Meteo and the indoor-climate modifiers into the WM calculation.
- Seed the catalogue with 25–30 plants.
- Soft launch to 20–30 plant-parent friends. Collect retention signal and advice-quality feedback.
- Iterate on weights based on feedback before any public launch.